Cyclic $n \times n$ Latin Squares representing solutions of (n, c)-Coins Problems for $0 \leq c \leq n$

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Abstract

(n, c)-Coins Problems [1] generalize n-Queens Problems allowing no more than c coins on any row, column, sum-diagonal or difference-diagonal of an $n \times n$ square instead of the usual 1 queen on any row, column, sum-diagonal and difference-diagonal.

We prove the following theorem :

For every positive natural number $n \in \mathbb{N}$ except $n \in \{2, 3, 4, 6\}$, there exists a Cyclic $n \times n$ Latin Square representing solutions of (n, c)-Coins Problems for all $c \in \mathbb{N}$ with $0 \leq c \leq n$.

We have an *existence proof* in `Lean` and corresponding *construction code* in `Haskell`, `Julia` and `Lean`.

See GitHub and also MO#510135, PSE#138939, MO#516225.

The proof uses the *Injective-Majorization Principle*. The existence of a single function on the whole Cyclic $n \times n$ Latin Square that is injective on each diagonal implies that for every threshold $0 \leq k < n$, the number of cells on each diagonal with value at most k is at most $k + 1$. The principle avoids intensive cardinality counting on each diagonal, which would computationally explode for large n.

The proof uses four subsets of $\mathbb{N}$ covering all n except $n \in \{8, 9, 15, 21\}$:

- $n \equiv 1, 5 \pmod 6$ (with $n \geq 1$),
- $n \equiv 2, 4 \pmod 6$ (with $n \geq 10$),
- $n \equiv 3 \pmod 6$ (with $n \geq 27$),
- $n \equiv 0 \pmod 6$ (with $n \geq 12$).

For all four subsets a Cyclic Latin Square can be constructed collectively for all their elements.

Constructing a Cyclic Latin Square for the four uncovered natural numbers $n \in \{8, 9, 15, 21\}$ is handled individually.

Constructing a Latin Square representing solutions of (n, c)-Coins Problems for natural numbers $n \in \{4, 6\}$, not dealt with by the theorem, is also handled individually.

1 Cyclic Latin Squares

1.1 From (n, c)-Coins Problems to Latin Squares

Given $n > 0$, and $c \in \{0, \ldots, n\}$ solutions of (n, c)-Coins Problems are subsets S_c of $n.c$ cells of an $n \times n$ square $S_{i,j}$ such that every row, column, s-diagonal ($i + j = s$), and d-diagonal ($i - j = d$), contains at most c cells of S_c.

Le, Li and Wang [1] construct each S_c *separately*.

We construct a square $S_{i,j}$, representing S_c as $\{(i, j) \mid S_{i,j} \leq c\}$, constructing all S_c for $0 \leq c \leq n$ *simultaneously*.

2 Theorem Statement

Theorem 2.1. *$\forall n \in \mathbb{N},\ n > 0,\ n \notin \{2, 3, 4, 6\}$, $\exists p$, a permutation of $\{0, \dots n-1\}$, such that $S_{i,j} = (i + p(j)) \bmod n$ is a Cyclic Latin Square satisfying :*

(i) $\forall i \in \{0, \dots n-1\}$, *the row map* $j \mapsto S_{i,j}$ *is bijective.*
(ii) $\forall j \in \{0, \dots n-1\}$, *the column map* $i \mapsto S_{i,j}$ *is bijective.*
(iii) $\forall k : 0 \le k < n, \forall s \in \mathbb{N}$, $|\{(i,j) \in s\text{-diagonal with } S_{i,j} \le k\}| \le k + 1$.
(iv) $\forall k : 0 \le k < n, \forall d \in \mathbb{Z}$, $|\{(i,j) \in d\text{-diagonal with } S_{i,j} \le k\}| \le k + 1$.

3 Injective Majorization

Instead of using direct cardinality counting for (iii) and (iv) above, which would computationally explode for large n, we use a principle that reduces the problem for *all* thresholds $0 \le k < n$ simultaneously to constructing a single dominated injection.

3.1 The Injective-Majorization Principle

Let $S_{i,j}$ with $0 \le i, j < n$ be a square and let D be any diagonal.

Lemma 3.1. *If there exists a function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that:*

(1) Majorization : $\forall i, j : 0 \le i, j < n, f(i,j) \le S_{i,j}$.
(2) Diagonal Injectivity : f *is injective on* D.

Then for every threshold $k \in \mathbb{N}$, the set $D_{\le k} = \{(i,j) \in D \mid S_{i,j} \le k\}$ has cardinality at most $k + 1$.

Proof. By condition (1), $0 \le f(i,j) \le S_{i,j} \le k$ so $f(i,j) \in \{0, \dots, k\}$. By condition (2), the restriction of f to $D_{\le k}$ is an injective map into $\{0, \dots, k\}$. Therefore $|D_{\le k}| \le |\{0, \dots, k\}| = k + 1$. □

4 Explanantion and Illustation of the Lean Proof

TBD (work in progress)

References

[1] Mao-Hua Le, Wen Li, and Edward T. Wang, *A generalization of the n-queen problem*, Journal of Systems Science and Mathematical Sciences **3** (1990), no. 2, 183–192, https://sysmath.cjoe.ac.cn/jssc/CN/Y1990/V3/I2/183.